Aseptic Technique Policy

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff undertaking procedures requiring aseptic technique

Contents

Introduction

Aseptic technique (AT) describes a set of infection control practices used to minimise the risk of contamination from pathogenic organisms and protect the patient from infection during invasive procedures.[1,2]

It is essential that all clinicians performing invasive procedures are competent in performing AT and can apply the principles across a range of procedures and settings.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☒
Other:	☐		

Definitions

Asepsis	Free from pathogenic organisms. The absence of microorganisms that can cause infection.[1,2,3]
Aseptic technique	To prevent pathogenic organisms, in sufficient quantity to cause infection, from being introduced to susceptible body sites by hands, surfaces and equipment.[4]
Aseptic Field	Controlled workspace that contains and protects procedure equipment from contamination, e.g., dressing pack, disinfected plastic trays, sterile procedure packs.[1,4]
Critical aseptic field	A controlled work area that ensures asepsis during procedures, e.g., central venous catheter insertion, large complex wound dressing, surgical procedure.[1,2,3]
General aseptic field	Controlled workspace that **promotes** asepsis, e.g., simple wound dressing.[1,2,3]
Micro critical aseptic field	A small critical aseptic field used to protect key parts or sites **within** a main critical or general aseptic field, e.g., syringe caps, sheathed needles, covers or packaging.[1,2,3.]
Key part	Sterile components of procedure equipment that must be protected from contamination during an aseptic procedure, e.g., needles, introducers, surgical instruments, invasive devices, bungs.[1,2,3,4]
Key site	Insertion site or access site or wound that is connected to or part of the patient, e.g., insertion/access sites of peripheral and central venous devices, urinary devices, open wounds.[1,2,3,4]

Policy Principles

The purpose of this policy is to:

- Provide a standardised, principle-based approach to AT by which clinicians can be educated, assessed and monitored to ensure compliance to AT principles
- Provide evidence-based practice guidelines that provide an expected standard for clinical staff performing AT, including locum or agency staff, at SCGH and OPH.

These principles apply to:

- Any procedure or insertion of medical device where there is entry to one or more of the body's normal defences such as skin, mucous membranes, or body cavity. This includes an interruption to a circuit or device
- The preparation and maintenance of sterile medicines
- The collection and processing of diagnostic samples to ensure accurate results inform patient treatment.[3,4]

Responsibilities

All clinicians who perform aseptic procedures have a responsibility to perform AT according to hospital policies, procedures and protocols.

Heads of Departments (HODs)
HODs are responsible for ensuring all clinical staff who perform aseptic procedures are trained and assessed as competent in AT.

New Employees and Clinical Staff who Perform AT Procedures
Aseptic Technique E-learning
All staff must complete the SCGOPH AT e-learning program:

- within six weeks of commencing employment AND
- three yearly as part of reassessment of AT competency.

AT e-learning should be completed within six weeks prior to undertaking an AT competency. Proof of completion of AT e-learning is required before being assessed for AT competency.

Aseptic Technique Competency
All SCGOPHCG staff, undertaking procedures requiring AT, must be assessed as competent in AT by a trained AT assessor, using the SCGOPHCG AT Competency tools:

- within six weeks of commencing employment, AND
- re-assessed at a minimum every three years.

The SCGOPHCG AT Competency Tools are available via HealthPoint on the Infection Prevention and Control AT page.

If a staff member does not meet the performance criteria, further training should be undertaken, and re-evaluation should occur at an agreed date as outlined by Aseptic Technique 'Not Yet Competent' document. This is available via HealthPoint/Infection Prevention and Control/Aseptic Technique.

Record Keeping
Record of training e-learning (ICATC 002) and competency (ICATC 003) are to be entered on LATTICE and maintained by each department.

Recognition of Prior Learning (RPL)
If AT training has occurred within the last 12 months in Australia and evidence can be provided, the new employee is not required to complete the SCGHOPHCG AT e-learning. However, the new employee must be deemed competent by a practical assessment using the SCGOPHCG AT Competency Tool within six weeks of employment.

Aseptic Technique Program

The Infection Prevention and Control (IP&C) Unit in conjunction with the Aseptic Technique and Invasive Devices Working Group (ATAID) are responsible for the implementation and evaluation of compliance with AT at SCGH and OPH. The working group will ensure the SCGOPHCG AT program aligns with the National Safety and Quality Health Service (NSQHS) Standards (Version 2): Preventing and Controlling Infections Standard, AT requirements.

Aseptic Technique Assessors
AT Assessors have been assessed as being competent to undertake auditing and competency assessments. To become an assessor clinical staff are required to:

- Complete the SCGH AT e-learning program
- Be deemed competent in AT assessment by a trained AT assessor
- Assess a staff member as AT competent under the supervision of an endorsed AT assessor

To maintain AT Assessor status the following must be achieved:

- Complete the SCGOPHCG AT eLearning program every 12 months
- Participate in the AT audit schedule
- View the “Overview of AT” video
- Attend the AT assessor refresher session every 12 months

Where specialty skills are being assessed (eg; infusaport insertion and PVC insertion), the AT assessor needs to be competent in these skills to oversee the competency assessment.
Assessor status will be withdrawn if annual eLearning and/or 3-yearly AT Competency assessment is not renewed within six months of date renewal is due.

Aseptic Technique Auditing
Compliance with AT will be determined via AT audits undertaken at a minimum of twice a year by clinical departments where AT procedures are performed. The departments should ensure a trained assessor is available to conduct the audits.

The Infection Prevention and Control Unit will co-ordinate the auditing schedule and will provide feedback of audit results to encourage ongoing improvement. The SCGOPHCG AT audit tools must be used and are available via HealthPoint/Infection Prevention and Control/Aseptic Technique.

Aseptic Technique Principles

AT is based on practices that aim to achieve asepsis for all clinical procedures by applying a set of principles that ensure key part and key site protection. The five essential principles of AT are:

- Asepsis is the aim for all invasive procedures, including the maintenance and use of invasive clinical devices. This is achieved by key part and key site protection from microorganisms transferred from the healthcare worker and the immediate environment[1,2]
- A risk assessment prior to performing procedures is required to identify key parts and key sites. This will determine whether a Standard or Surgical AT is indicated that the correct infection prevention measures are in place and the correct equipment available to perform the procedure safely[2,3,4]
- Non-touch technique should be the goal for all procedures. If it is necessary to touch key parts or key sites directly, sterile gloves are required[1,2,3]
- AT should be performed using a logical sequence to maintain asepsis. Refer to Appendix 1: Sequencing of procedures requiring standard aseptic technique and Appendix 2: Sequencing of procedures requiring surgical aseptic technique.
- Patient and healthcare worker safety during aseptic procedures is supported by the application of standard precautions.[1]

Application of AT principles to AT practice

Environmental controls

Prior to conducting a procedure ensure nearby avoidable environmental risks are managed. Examples of environmental risks include aseptic field less than one metre from handwash basin, bed making, cleaning of the environment in proximity, use of commodes by other patients in shared rooms.

Hand hygiene

- Hand hygiene (HH) must be performed according to the “The 5 moments for hand hygiene.”
- Additional critical moments before, during and after a procedure when HH should be performed include:
  - Before and after collecting equipment
  - After setting up an aseptic field
  - Immediately before donning gloves
  - Immediately after completing the procedure and removing gloves
  - Immediately after cleaning up and disposing of equipment and waste
- Routine HH should be performed using non-medicated neutral pH soap/foam hand wash and running water or alcohol-based hand rub.
- Where full barrier precautions are required, a surgical hand scrub using an approved antimicrobial skin cleanser or waterless hand rub formulation is required.[1,2,3] Refer to specific SCGH and OPH Operating Suite / Theatre policies.

Glove use

- Non-sterile gloves during a Standard AT procedure may be indicated to protect from exposure to blood, body fluids or toxic drugs during the procedure
- Sterile gloves are required for all Surgical AT procedures and any Standard AT procedure where a non-touch technique cannot be achieved. The use of sterile gloves when touching key parts and/or key sites is used to minimise the risk of contamination.

- A healthcare worker should assess their own competency and skills in performing a procedure to determine whether there is a requirement/risk that key parts/sites will be touched. If contact with the key parts/sites is required, then the use of sterile gloves is necessary.[1,2,3]

Use of other PPE

- The use of other PPE i.e., aprons or facial protection, should be according to standard precautions and based on the risk assessment of exposure to blood and body fluids
- Maximum barrier precautions may be required during invasive procedures to further reduce the risk of contamination, e.g., as required for insertion of CVC.[1,2,3]

Risk Assessment and selection and management of the aseptic field

Although the principles of AT are applied to all invasive procedures, the level of practice changes depending on a risk assessment. The risk assessment will determine:

- Type and complexity of the procedure
- The key parts and key sites
- Whether key parts and key sites need to be touched
- The appropriate infection prevention and control measures to protect key parts and key sites

Types of aseptic technique

	STANDARD AT Promotes asepsis	**SURGICAL AT** Ensure asepsis
PROCEDURE	• Technically simple • Short duration less than 20 min • Few key sites.	• Technically complex • Takes greater than 20min • Large open key sites.
ASEPTIC FIELD	Use a general aseptic field and/or critical micro aseptic field  Figure 1. General Aseptic Field. The general aseptic field, demonstrated by the yellow outline in Figure 1., promotes asepsis with the mandatory use of Micro Critical Aseptic Fields (red outline in Figure 1.) and non-touch technique.[1,2]	Use a critical aseptic field and critical micro aseptic field.  Figure 2: Critical Aseptic Field. Adopted from ANTT. Figure 2. Identifies the critical aseptic field with the red circle outline demonstrating the clinician use of sterile cloves and a main critical sterile field with many Micro Critical Aseptic Fields within this.
Personal Protective Equipment (PPE)	• Nonsterile gloves to remove dressing. or risk of exposure to blood or body fluids. • Sterile gloves if a key part or key sites must be touched or forceps not used. • Apron/face protection as per Standard Precautions.	• Sterile gloves • Often, full barrier precautions – sterile gown, mask, hair covering, sterile drapes.
ENVIRONMENT	• Work surface must be cleaned with combined detergent / disinfectant wipe and allowed to air dry before and after the procedure. For example: dressing trolley • Bed making or cleaning activities in close proximity should be avoided. • Aseptic field to be over one metre from hand wash basins to avoid splashing	• Work area and surfaces cleaned with combined detergent/ disinfectant wipe before and after procedure. Staff activity strictly controlled • Environmental risk removed or avoided.

Non-Touch Technique

Non-touch technique is always required to maintain asepsis. This is a technique whereby the healthcare workers hands do not touch, and thereby contaminate key parts and key sites. Asepsis can be achieved by either:

- Using a non-touch technique e.g. sterile forceps or sterile gauze
- Using sterile gloves. If it is necessary to touch key parts or key sites directly, sterile gloves are used to minimise the risk of contamination. Even when using sterile gloves, key parts and key sites should not be touched unless necessary to do so.[2,4]

Related Documents

- [Australian guidelines for the prevention and control of infection in healthcare]
- [National Safety and Quality Health Service Standards]
- [SCGOPHCG Infection Prevention and Control Manual:]
  - Cleaning of Patient Care Equipment
  - Environmental Cleaning Policy
  - Hand hygiene Policy
  - Safe injection Practices Policy
  - Specimen collection for Microbiological examination Guideline
  - Storage of sterile and non-sterile stock in clinical areas
- [SCGH Practice Guidelines for procedures requiring AT]
- [SCGOPCG AT Competency Assessment Tools]
- [SCGOPCH AT Audit Tools]

References

1. National Health and Medical Research Council. Australian Guidelines for the Prevention and Control of Infection in Healthcare [Internet]. Canberra; 2019 [cited 2022 August 4]. Available from: https://app.magicapp.org/#/guideline/6465
2. The Association for Safe Aseptic Practice. ANTT Clinical Practice Framework version 4.0. London, The Association for Safe Aseptic Technique Practice, United Kingdom, 2015.
3. Australian Commission on Safety and Quality in Health Care. Principles for aseptic technique: Information for healthcare workers [Internet]. Australia [cited 2022 August 4]. Available from https://www.safetyandquality.gov.au/sites/default/files/2022-01/principles_for_aseptic_technique-_information_for_healthcare_workers_-_december_2021.pdf
4. Government of South Australia. Aseptic Technique Staff training, self-assessment and competency workbook [Internet]. Adelaide, Government of South Australia, 2022 November [cited 2022 August 4]. Available from https://www.sahealth.sa.gov.au/wps/wcm/connect/public+content/sa+health+internet/clinical+resources/clinical+programs+and+practice+guidelines/infection+and+injury+management/healthcare+associated+infections/prevention+and+management+of+infections+in+healthcare+settings/aseptic+technique#:~:text=Aseptic%20technique%20aims%20to%20prevent,of%20staff%2C%20surfaces%20or%20equipment.
5. Australian College for Infection Prevention and Control. Aseptic Technique Resources [Internet]. Hobart [cited 2022 August 4]. Available from https://www.acipc.org.au/resources/aseptic-technique-resources/
6. The Joint Commission. Preventing Central Line-Associated Bloodstream Infections: Useful Tools [Internet]. An International Perspective. 2013, November 20 [cited 2022 August 8]. Available from clabsi_toolkit_tool_3-21_scrub_the_hubpdf.pdf (jointcommission.org)

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Appendix 1: Sequencing of procedures requiring standard aseptic technique.

Adapted from ACIPC AT implementation toolkit.[5]

Risk assess and confirm standard aseptic technique appropriate for procedure.
• The procedure is technically simple • Short in duration (less than 20 minutes) • Clinician is experienced / competent at performing procedure • Involves few and small key parts and key sites • Key parts or sites will not be touched.
Apply environmental controls
Ensure that there are no avoidable nearby environmental risk factors. This may include, but is not limited to waste management, cleaning of the nearby environment, bed making, patient use of commode nearby, privacy screens across work areas.
Infection prevention and control components
• Hand hygiene (HH) • Gloves – Non-sterile gloves are typically used to protect from potential exposure to body fluids or harmful substances. If it is likely key parts or key sites need to be touched directly, sterile gloves MUST be used. • Personal Protective Equipment • General Aseptic Field
Prepare for procedure
1. Perform HH. **2.** Clean the tray/trolley/work surface with combined detergent/disinfectant wipe. Allow to dry. **3.** Identify and gather equipment for procedure. (Inspect packaging for damage; check sterility indicators (where applicable) & expiry dates **4.** If necessary, move to where the procedure will take place **5.** Perform HH **6.** Prepare general aseptic field. Open equipment using non-touch technique **7.** Position and prepare the patient, using non-sterile gloves where appropriate to protect from exposure to potential body fluids harmful substances e.g., removing soiled dressing.
Perform the procedure
8. Once ready to commence procedure and required equipment prepared, remove gloves (if used in preparation for procedure) and perform HH **9.** Apply gloves as appropriate **10.** Perform the procedure using non-touch technique, ensuring all key parts/components are protected at all times. Sterile items must only be used once and disposed into waste bag. Only sterile items may come in contact key sites and sterile items must not come into contact with non-sterile items.
Waste management and cleaning of equipment
11. On completion of the procedure the clinician should remove their gloves and PPE and perform HH. Dispose of all waste. Clean/reprocess equipment/worksurfaces as appropriate. Perform HH.

Appendix 2: Sequencing of procedures requiring surgical aseptic technique.

Adapted from ACIPC AT implementation toolkit.[5]

Risk assess and confirm standard aseptic technique appropriate for procedure.
• The procedure is technically complex • Clinician experience / competency requires touching of key sites/parts • Long in duration (longer than 20 minutes) • Involves large open key sites or large or numerous key parts • Key parts or key sites need to be touched as part of procedure
Apply environmental controls
Ensure that there are no avoidable nearby environmental risk factors. This may include, but is not limited to: waste management, cleaning of the nearby environment, bed making, patient use of commode nearby, privacy screens across work areas.
Infection prevention and control components
• Hand hygiene (HH) – surgical hand scrub • Sterile gloves • Personal Protective Equipment • Critical aseptic field • Maximum barrier precautions
Prepare for procedure
1. Apply PPE as required (e.g., cap & surgical mask to protect aseptic field) **2.** Perform HH **3.** Clean the tray/trolley/work surface with combined detergent/disinfectant wipe. Allow to dry **4.** Identify and gather equipment for procedure. (Inspect packaging for damage; check sterility indicators & expiry dates and ensure any additional equipment, is clean) **5.** If necessary, move to where the procedure will take place **6.** Perform HH **7.** Prepare critical aseptic field and open equipment onto critical aseptic field using non-touch technique **8.** Position and prepare the patient using gloves where appropriate to protect from exposure to potential body fluid or harmful substances
Perform the procedure
9. Once ready to commence the procedure and the required equipment is prepared, remove gloves (if used in preparation for procedure) and perform surgical scrub (as required for the procedure) **10.** Apply sterile gown and gloves **11.** If required add sterile drapes as necessary **12.** Perform the procedure using non touch technique; ensure all key parts/components are always protected. Sterile items must only be used once and disposed into waste bag. Only sterile items may come in contact with key sites and sterile items must not come into contact with non-sterile items.
Waste management and cleaning of equipment
On completion of the procedure the clinician should remove their gloves, PPE and perform HH. Dispose of all waste. Clean/reprocess equipment/worksurfaces as appropriate. Perform HH.